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ORIGINAL RESEARCH

Evaluation of EU-27 countries as manufacturing location: strengths and weaknesses model

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Abstract
In the recent years there has been an increased interest towards relocating manufacturing back to Europe. Consequently, it becomes important to develop evaluation models of manufacturing location. We use secondary data to establish model for European Union 27 (EU-27) countries. Findings reveal in pairwise model comparison, that larger European countries seem to be the best locations in general (France, Germany, Italy, and Spain), however, few smaller countries perform high as well (like Ireland, Sweden, Luxemburg and Finland). In general, small member countries have the lowest performance. Typically, these are weak in population and severance pay terms. Lower-performing countries often also miss the larger role of medical and pharmaceutical industry within exports. Results were compared to classical rank-based evaluation. We did find some differences, and it could be said that larger countries lost their higher status, but Ireland and Sweden were still in highest ranks.

Keywords Manufacturing relocation · Reshoring · Global supply chains · European Union · International business

1 Introduction

In the recent years, bringing manufacturing back to developed economies has become an increasingly prominent topic in both academic and industry discussions since the early 2010's (Ellram et al. 2013; Ancarani et al. 2015). This is often taking place in the context of global supply chain disruptions and geopolitical uncertainties (Hilmola 2023; Zighan et al. 2023) as well as in the increased role of nation states and economic blocks in industrial sector (through regulation and policies; see Mariotti 2024). This trend, which is often attributed to the pursuit of supply chain resilience, competitive advantage, and closer proximity to markets (Benstead et al. 2017; Zighan and Dwaikat 2023). In Europe, reshoring has been accelerated by multiple factors, including the coronavirus era, highlighted the weakness of global supply chains, and companies have learnt their lessons in a costly way, due to increasing freight rates, availability issues and increased inventory holdings (Free and Hecimovic 2021; Boehme et al. 2021; Zighan et al. 2023). Following the Russian invasion of Ukraine, which has rendered some Eastern European manufacturing hubs unstable and unreliable (Hilmola 2023). The automotive industry, with its shift toward electric vehicles, exemplifies these shifts and the EU's increasing drive to reinvigorate regional manufacturing capacity (Razmjoo et al. 2022).

In addition, technological development allows more distributed operations, which still can be run efficiently with the ability to introduce new products fast to markets, provide short lead times and customization (Purvis et al. 2021). However, it should be noted that reshoring back to developed economies is not entirely problem-free as cost issues are constantly a threat (if the supply chain is also more based there; Günay et al. 2019), and getting used to manufacturing

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tensions is a challenge for all parties, like customers, manufacturer, and suppliers (Harper 2022). However, developed economies might provide a better overall quality level and framework for manufacturing companies to operate (Engström et al. 2018b). This reshoring development has been supported since global financial crisis (post year 2008) by more restrictive and regulatory policies of developed nations, while liberalization policies have meanwhile been hurt (Mariotti 2024).

Within this changing landscape, the debate around reshoring has also highlighted the need for comprehensive models that can assess and compare the suitability of different regions or countries as manufacturing locations. Key drivers in this evaluation include labour costs, proximity to markets, regulatory environments, and the availability of resources and technology. The European Union, with its 27 diverse member states, offers a unique context in which manufacturing favourability can vary significantly depending on these criteria. The impact of EU policies, such as those promoting sustainability (Castillo-Díaz et al. 2024) and technological innovation, further complicates these location decisions (Engström et al. 2018a, b). However, the relatively high operational costs in developed economies, coupled with the pressures on companies to remain agile and responsive to market demands, continue to pose challenges for reshoring efforts (Günay et al. 2019).

Using Analytic Hierarchy Process (AHP) in the evaluation and decisions of different branches has been popular since its introduction more than four decades ago. Emrouznejad and Marra (2017) identified in their literature review 8441 published articles in the period of 1979–2017. Interestingly, only very few papers out of the large pool of AHP articles concerned location problems (in total three). One of these research works dealt with hospital location, which is out of the scope of this research, however, one was from textile manufacturing location evaluation (Ertuğrul and Karakaşoğlu 2008). Factors that were used in the evaluation model as evaluation criteria include labour climate, proximity to markets, community considerations (cost and availability of utilities, taxes, regulation, development support), quality of life, and proximity to suppliers and resources. The third study identified was on location criteria concerned warehouse location decision (Dey et al. 2017), with evaluation model criteria such as transportation capacity and modes, space availability and costs. Chou and Yu (2013) also proposed an evaluation model for distribution centers, which was rather similar to these former, but was more detailed and deeper in its evaluation measures, and it also incorporated investment conditions (such as tax issues, restrictions and land availability). Sequeira et al. (2021) used AHP in manufacturing reshoring decisions at the company level (also one type of location decision), where cost was one important issue together with other criteria like quality, time, flexibility (Johansson and Olhager 2018), innovation, and sustainability (similar emphasis on quality as supplier selection model of Rodrigues et al. 2022). Panova and Hilmola (2018) adopted a macro-economic approach in manufacturing location selection and used logistical (proximity to key markets and availability of e.g., seaport) and economic measures (like price of electricity and energy) in the evaluation too.

To address the research gaps, this paper develops a model to compare the strengths and weaknesses of EU member countries, to answer the following Research Questions (RQs):

RQ1: What are the key criteria that determine manufacturing favourability across EU member countries?

RQ2: How do these criteria vary among EU-27 countries, and what implications does this have for companies considering production relocation within Europe?

By answering these questions, we aim to provide valuable insights into the strengths and weaknesses of manufacturing in EU countries; to better understand, how they differ in manufacturing favorability. This study is relevant not only for macro-economic analysis of foreign direct investments and trade, but also for companies and practitioners considering relocation. Furthermore, new product groups are located in Europe (such as electrical vehicle components).

In the rest of the paper, research proceeds as follows: In Sect. Literature review of reshoring is provided literature review of reshoring of manufacturing back to developed economies, and particularly processes concerning Europe. Evaluation model and used data are introduced in Sect. Used evaluation model and data. Thereafter, in Sect. Evaluation model with strengths and weaknesses: results of EU-27 countries we provide pairwise analysis results of European Union 27 (EU-27) countries. Analysis provides further dive in the strengths and weaknesses of each country together with the consideration of key export sectors. Findings are compared to classical rank-based comparison in Sect. Rank-based evaluation. We discuss research findings in Sect. 6 and conclude research work as well. Article ends with a proposal of future research avenues in this area.

2 Literature review of reshoring

Finding the right manufacturing locations constitutes a vital part of the manufacturing strategy implementation (Mohiuddin et al. 2019; Li et al. 2024; Uluskan et al. 2017). The location decision must be aligned with the overall business strategy and enable the firm to gain competitive advantages in resources, markets, efficiency, or strategy assets (Dunning 1988, 1998). The most common manufacturing location

strategy in recent decades has been offshoring, where firms locate manufacturing in distant low-cost locations (Boffelli and Johansson, 2020; Grappi et al. 2020; Johansson and Olhager 2018; Nujen et al. 2018). A more recent manufacturing location strategy is reshoring, where firms instead locate manufacturing in proximity to the market (Dachs et al. 2019; Fel and Griette 2017; Hilletofth et al. 2019, 2021). The reshoring manufacturing location strategy has received significant attention in recent years by scholars, practitioners, and policymakers (Barbieri et al. 2018; Di Mauro et al. 2018; Dachs et al. 2019; Nujen et al. 2019). The research topic is in its infancy, but is growing very rapidly (Fratocchi and DiStefano, 2019; Li et al. 2024; Tate et al., 2014).

A large share of the available reshoring strategy literature has a European perspective (Eriksson et al. 2021; Hilletofth et al. 2019; Lampón and González-Benito 2020; Panova and Hilletofth 2017) and includes three main research streams: reshoring decision-making (e.g., Wiesmann et al. 2017; Engström et al. 2018a, b; Sirilertsuwan et al., 2019; Boffelli and Johansson, 2020), reshoring implementation (e.g., Zhao and Huchzermeier 2017; Boffelli et al. 2018; Lund and Steen 2020) and reshoring outcomes (e.g., Uluskan et al. 2017; Stentoft et al. 2018; Johansson et al. 2019). The decision-making stream has by far received most attention (Barbieri et al. 2018; Hilletofth et al. 2019; Stentoft et al. 2016). The majority of research in this stream has focused on identifying and understanding the factors influencing the decision-making, that is, the decision variables/factors/criteria (e.g., Ellram, 2013; Foerstl et al. 2016; Wiesmann et al. 2017; Fratocchi and Di Stefano 2019; Johansson et al. 2019) and the role the factors may have in the decision-making, for instance, constituting a driver, barrier, or enabler (Foerstl et al. 2016; Engström et al. 2018a; Sirilertsuwan et al., 2019). Understanding the decision factors is critical to make well-informed decisions (Li et al. 2024).

Many decision factors have been proposed and they are typically structured into decision content frameworks, occasionally using a specific theoretical lens, such as Dunning's eclectic paradigm (Dunning, 1973, 1988, 1998) of cost-seeking, market-seeking, resource-seeking and strategic-asset seeking (Ancarani et al. 2015; Ellram et al. 2013). The categorization used to classify decision factors varies. In one research work, the factors are divided according to the main goal (cost-efficiency or perceived-value) and the level of analysis (internal or external) (Fratocchi et al. 2016). In another one, the factors are divided into global dynamics, host country, home country, supply chain and company-specific issues (Wiesmann et al. 2017). In other frameworks, the factors are grouped based on competitive priorities, arguing that firms relocate to increase competitiveness (Benstead et al. 2017; Srai and Ané 2016). Scholars argue that previous decision content frameworks have not been as robust as needed (Bailey and De Propris 2014; Stentoft et al. 2015) and calls for more inclusive frameworks.

According to Dunning's eclectic paradigm (Dunning, 1973, 1988, 1998), companies may relocate their manufacturing to access resources that are not available or easier to access in a particular location, which are called resource-seeking factors. In this category, labour related factors have intensively been discussed by scholars within the EU manufacturing context. For instance, such are motivated, experienced and stable labour resources (Boffelli et al. 2018; Rainnie 2021), labour cost (Ancarani et al. 2021; Dachs et al. 2019), and labour productivity (Panova and Hilletofth 2017; Theyel and Hofmann 2021). Factors in the market-seeking category allow companies to exploit market opportunities in a particular location. In this category, Dunning (1979) noted macro-economic and macro-organizational policies shape the industrial climate, subsequently influencing firms' operations. Castillo-Díaz et al. (2024) have also noted that economy plays a significant role in achieving several sustainable development goals. Another important factor related to macro-economics that affects companies in reversing the offshoring decision is inflation, which raises production costs and undermines profitability (Piatanesi and Arauzo-Carod, 2019; Stentoft et al. 2016). Efficiency seeking factors help to achieve cost-efficient and productive manufacturing in a particular location. For example, the stability of the national economy can affect companies in making decisions to a large extent, it is perceived as one source of competitiveness for companies' strategic realizations (Bailey et al. 2018). Under the national economy category, we can find factors such as population, size of the country, and logistics performance (Rainnie 2021; Wan et al. 2019). These four factors (macro-economy, labour, inflation and national economy) will be discussed in the following section, as they constitute the used evaluation model.

3 Used evaluation model and data

Manufacturing location decision framework is rather general in this research, and it is not dependent or based on any specific sub-industry and its special requirements. Framework decision variables are divided in four different main themes (see Fig. 1): (1) Macro-Economy, (2) Labour, (3) Inflation and (4) National Economy. In overall, framework consists of fourteen decision variables, and analysis is based on two earlier research works, and methodology developed further in these (it differs slightly from original AHP, called as general choice algorithm (GCA), and it can compare quantitative second-hand data; see Panova and Hilmola 2018; Panova et al. 2019). The applied framework and its measures were disseminated for practitioners; CEOs and

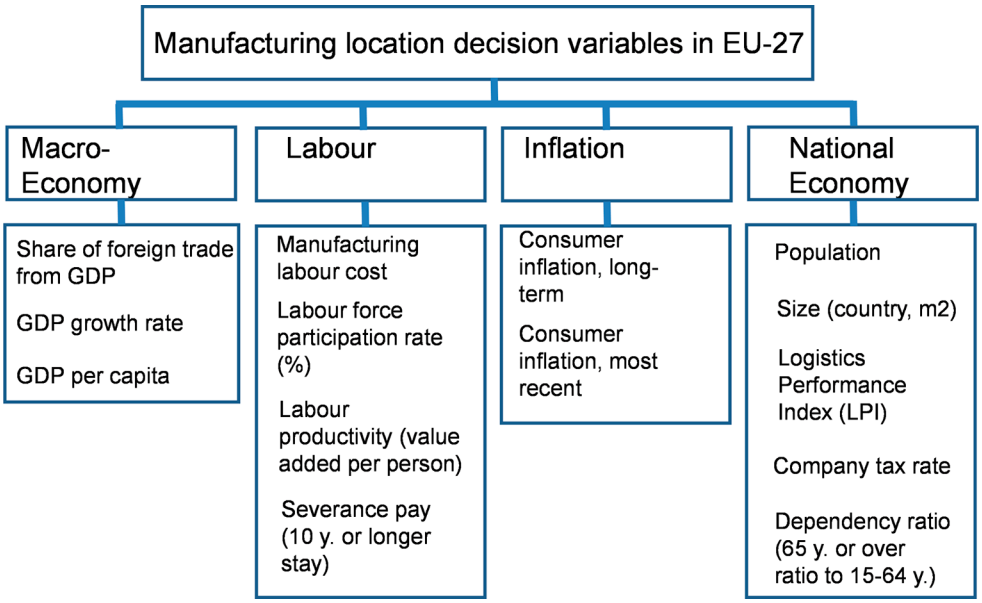


Fig. 1 Manufacturing location decision variables in used the model

senior managers of European manufacturing companies in a joint-seminar held in Gävle, Sweden in February 2023 (all of these companies had significant manufacturing operations in Sweden), and in follow-up seminar during September 2023. There was a high interest for this evaluation and the measures were considered as interesting and relevant for manufacturing location decisions (at macro-level) from practitioner perspective. Based on the request and discussions from first seminar, three further measures were added (labour productivity, severance pay and dependency ratio). These all were felt to be very important for industrial companies making their location decision (as it is a long-term and strategic decision; e.g., Joubioux and Vanpoucke 2016). Original framework consisted of eleven measures, which is mostly based on earlier research of manufacturing location decision in one larger country (Panova and Hilmola 2018), but also in parts to one large country evaluation concerning e-commerce operations location (Panova et al. 2019). Even if this used model with 14 different variables might sound complex, it was felt among industrial participants of seminar as interesting and useful as in the current complex operating environments decisions are not based on one or few measures alone.

In contrary to earlier models, we have included inflation in the model with two different measures. This is due to the reason that Europe has faced a rather big challenge with inflation in 2021-22, where in a few countries consumer inflation increased up to 20% and above. The model includes inflation as an average measure from ten-year period of 2012–2021. Together with this, the model uses annual inflation, which is gathered from these countries out of December 2022 (representing new inflation era with Ukrainian war effects). There is a great contrast between these two data sets of inflation as in ten-year average data highest inflation was 2.45%, and in the most recent it was 24.5% (both from Hungary). The lowest inflation in ten-year period was negative (-0.13%, Greece), where lowest level in most recent data was 5.4% (Luxemburg). It could be said that inflation has been sudden new challenge, and manufacturing companies have not dealt it within EU-27 area that much. It can be noted that some measures are negative in dataset, so they were rescaled to make them positive, where lowest performance was very low positive one (otherwise pairwise comparison would not have been possible). In practice this meant that all values of different countries were increased by 1% plus percent amount of Greece being in the negative territory (country having lowest performance in both rescaled measures). Therefore, for GDP growth all countries had additional 1.57% and for inflation 1.13%. It is very typical to rescale in pairwise comparisons used measures as e.g., in AHP zero and negative values can not be used (Millet and Schoner 2005). We also used a normalized max-min approach for these two variables (e.g., Vafaei et al. 2020) – differences to now used were rather small, and therefore the first mentioned was decided to be used in this research. As the framework was disseminated for the practitioners of manufacturing companies, inflation was a current issue that raised attention. In 2023 inflation was predicted to continue to be high, and this was somewhat challenging in terms of salaries increasing and higher cost of basic raw materials (like steel). Companies tackled these with productivity initiatives and investments.

From the data we could draw the conclusion that EU-27 countries are not that homogenous group as could be assumed in the first place. For example, highest share of trade from GDP (351.27%; average from 2012 to 2021) and

GDP per capita (115918.70 USD per person; average from 2012 to 2021) exists in Luxemburg as lowest share of trade from GDP was detected to be in Italy (57.47%) and lowest GDP per capita in Bulgaria (8770.40 USD per capita). GDP growth rate in European countries has been in observed decade (2012–2021) rather low, on the average 2.09%, but there are countries, which perform well above this (Ireland, 7.89%) as well as having negative growth rate (Greece, -0.57% and Italy, -0.2%). Situation is similar in other measures, like manufacturing labour cost, population, size of country, labour productivity, severance pay and company tax rates (see Table 1 for details). Only Logistics Performance Index (LPI) is having rather consistent performance throughout EU-27 countries – where highest performance (also in world, year 2018) is that of Germany (4.2), and lowest could be detected in Latvia (2.81). In addition, labour force participation rate does not differ that much among countries as does not inflation. Even if in some measures differences (relative) are not that great, but their relevance in practice is significant. In joint-seminar companies saw difficulties to acquire labour for Swedish factories (which is having highest labour force participation rate from Table 1).

Evaluation model uses pairwise comparisons (like AHP), and in these all-potential manufacturing locations (EU-27) are compared to each other (making a 27 times 27 matrix). All used measures had equal weight and effect on the overall scoring of countries (in both approaches used, general choice algorithm and rank-based evaluation). Question of course arises, whether used evaluation criteria variable should be seen as being better performance as showing lower amounts (e.g., labour costs) or are higher values preferable (like in LPI). In the evaluation model of this research work following items were seen as being better and having higher value: foreign trade from GDP, GDP growth, GDP per capita, population, surface area, LPI index and labour productivity. In turn following variables are better, when they show lower value: labour force participation rate, inflation, labour costs (LCI), company tax rate, recent inflation rate, severance pay and dependency ratio. It is of course debatable whether e.g., higher GDP per capita will

Table 1 Used fourteen variables and their performance in different EU-27 countries. Source (data): Eurostat (2023), World Bank (2023a), United Nations (2023)

Country	Foreign trade from GDP	GDP growth (annual)	GDP per capita	Labour force participation rate	Inflation	Recent inflation	Population	Surface area	Labour cost for LCI	LPI index	Tax rate	Labour productivity	Severance pay	Dependency ratio
Austria	104.8	0.9	49199.9	75.7	1.8	10.2	8978929	83858	37.1	4.0	25.0	85.4	0.0	44.0
Belgium	161.5	1.3	45736.0	67.7	1.6	10.4	11631136	30528	41.1	4.0	25.0	119.7	0.0	41.9
Bulgaria	124.8	2.1	8770.4	69.2	1.4	16.9	6838937	110912	6.5	3.0	10.0	16.1	4.3	45.7
Croatia	92.3	1.7	14172.0	65.6	1.0	13.1	3879074	56538	10.8	3.1	18.0	24.2	14.4	46.3
Cyprus	142.3	1.7	27736.2	72.8	0.1	7.9	904705	9251	17.3	3.2	12.5	37.3	0.0	32.7
Czechia	147.2	1.9	21383.9	73.7	2.1	15.8	10516707	78866	14.1	3.7	19.0	32.2	13.0	40.7
Denmark	104.9	1.8	59788.5	77.5	0.9	8.7	5873420	43094	45.3	4.0	22.0	126.1	0.0	42.7
Estonia	153.1	3.3	21098.5	76.6	2.0	17.6	1331796	45100	13.6	3.3	20.0	32.1	4.3	42.2
Finland	75.6	0.7	48239.9	75.8	1.1	9.1	5548241	338145	33.8	4.0	20.0	84.3	0.0	44.7
France	61.2	0.9	40380.6	71.1	1.0	5.9	67842582	551500	37.4	3.8	26.5	70.9	10.8	46.7
Germany	86.0	1.0	45878.0	77.8	1.4	8.6	83237124	357022	36.7	4.2	30.0	77.4	21.7	47.7
Greece	71.6	-0.6	19666.0	68.0	-0.1	7.2	10603810	131957	16.9	3.2	22.0	29.6	26.0	52.8
Hungary	163.8	2.8	14949.4	67.2	2.4	24.5	9689010	93032	9.9	3.4	9.0	33.2	13.0	39.4
Ireland	219.3	7.9	69693.5	72.3	0.6	8.2	5060005	70273	32.4	3.5	12.5	522.5	21.0	33.7
Italy	57.5	-0.2	33565.2	64.1	0.9	11.6	58983122	301318	29.7	3.7	24.0	60.0	0.0	56.5
Latvia	123.7	2.9	16357.6	75.5	1.5	20.8	1875757	64600	10.5	2.8	20.0	24.8	13.0	48.8
Lithuania	146.4	3.4	17565.0	74.2	1.9	21.7	2805998	65300	10.1	3.0	15.0	27.1	8.7	50.8
Luxembourg	351.3	2.4	115918.7	69.7	1.4	5.4	645397	2586	42.2	3.6	24.9	83.9	8.7	34.6
Malta	293.5	5.5	27865.3	67.7	1.2	7.3	520971	316	16.8	2.8	35.0	42.0	0.0	33.7
Netherlands	152.8	1.3	51052.9	79.4	1.7	9.6	17590672	41528	37.4	4.0	25.8	104.8	14.3	44.9
Poland	98.3	3.4	14452.7	68.2	1.8	16.6	37654247	312685	11.0	3.5	19.0	30.2	13.0	40.2
Portugal	81.4	0.6	21875.6	74.0	0.8	9.6	10352042	91982	15.7	3.6	21.0	29.9	17.1	54.1
Romania	83.4	3.2	11092.7	65.9	2.4	16.4	19038098	238391	8.2	3.1	16.0	19.1	0.0	44.4
Slovakia	181.7	2.1	18484.1	70.7	1.6	15.4	5434712	49033	13.4	3.0	21.0	27.9	13.0	39.0
Slovenia	150.9	2.1	24345.1	72.3	1.1	10.3	2107180	20256	19.9	3.3	19.0	43.2	10.8	46.7
Spain	64.4	0.5	28441.5	74.2	1.0	5.7	47432805	505992	23.0	3.8	25.0	55.9	28.6	49.4
Sweden	85.0	1.9	55688.2	81.4	1.0	12.3	10452326	449964	37.3	4.1	20.6	91.5	0.0	37.3
Average	132.5	2.1	34199.9	72.2	1.3	12.1	16549215	153482	23.3	3.5	20.7	71.5	9.5	43.8
Min	57.5	-0.6	8770.4	64.1	-0.1	5.4	520971	316	6.5	2.8	9.0	16.1	0.0	32.7
Max	351.3	7.9	115918.7	81.4	2.4	24.5	83237124	551500	45.3	4.2	35.0	522.5	28.6	56.5

lead to a preferable manufacturing location as a country. There exist examples even among EU-27 countries, where manufacturing is prospering despite having high GDP per capita. For example, in Ireland GDP per capita is 69693.5 USD (second highest among EU-27 countries; Table 1). Irish manufacturing jobs have increased from 2011 to 2020 by 55,017 (30.4% increase) and manufacturing value added has increased in the same period by 240.7% (both statistics by Eurostat 2023). Irish balance of trade has been positive for three decades (and surplus is growing), and inward foreign direct investment to manufacturing sector of this country shows continuous growth too (OECD 2023). Similar performance could be detected from such high GDP per capita countries as Denmark and Netherlands too (with only exception that manufacturing jobs are on decline or show only slight growth, however, trade surplus, manufacturing value added, and exports are increasing). However, there of course exist countries in EU-27, which are having high GDP per capita, but have shown lower performance in the previous decade (like Finland and Sweden).

4 Evaluation model with strengths and weaknesses: results of EU-27 countries

The evaluation results with applied model are shown in Table 2. In general, it could be said that large countries do take most of the top positions (like France, Germany, Italy, and Spain) and smaller countries are least favourable manufacturing location. However, being a small country, does not necessarily mean that performance could not be high. For example, Ireland, Sweden, Luxemburg, and Finland are rather high in the evaluation, although being rather small in population and geographical size. It is also notable that differences in top positions as well as in the bottom are rather small (e.g., Ireland is 2.18 times higher performance than Latvia). Therefore, instead of ranks, Table 2 can be analyzed based on groups where countries seem to be belonging into.

To understand results better, was performance of each country analyzed further with spreadsheet concerning its performance profile. For highest performing countries of Table 3, these most valuable variables (strengths) were population (France, Germany, and Italy) and geographical size (Spain). As a weakness of most of these large countries was severance pay (France, Germany, and Spain) and GDP growth (Italy). These strengths were having highest co-efficient in the country profile and weaknesses were the lowest value. Interestingly, top export items of France and Germany (in 2020) were rather similar – medical industry products and complex mechanical products. In Spain these were automobiles (complex mechanical products) and petroleum products. As in Italy medical industry together with petroleum products played important role in exports.

Although France is the largest country in geographical size and Germany is the fourth largest among EU-27 countries, this was not considered as the main strength. Solution could be explained with the nature of pairwise comparisons – in surface area difference of France to Spain is rather small, and in addition, difference to other large countries is rather small as compared to France's population difference to other countries. Co-efficient of France in surface area is just lower than that of population (Table 2: 0.133 vs. 0.152). Same applies to Germany, but with an exception that it is the largest in the population among all EU-27 countries, and it is easier to understand, why population is seen as its strength (co-efficient in Table 1 is significantly larger). However, Germany is also the highest performing in LPI index, but as noted earlier, differences between countries are not that wide in this variable, and therefore, it is natural that this is not its main strength (co-efficient is so small in Table 2). Spain is having similar situation with these two countries, where its population is the fourth largest in Europe and most recent inflation is the second lowest. Within surface area its performance relative to these two is best, and having highest co-efficient, and therefore it was selected as the main strength by the linear programming model.

Situation in high performing smaller countries is different from earlier analyzed larger countries. Strength in Ireland (Labour productivity; see Table 3), Sweden (Surface area), Luxemburg (GDP per capita) and Finland (Severance pay) is not the same, and they are all somewhat different, and they do differ from larger countries. Weaknesses of Ireland (Severance pay), Sweden (Labour cost for LCI), Luxemburg (Surface area) and Finland (Population) do also differ, and do not hold any similar pattern, and difference remains to larger countries. In Ireland most important export products are animal and microbial related together with medical products. As in Sweden medical products also hold importance together with petroleum products. In Luxemburg (mostly steel products) and Finland (petroleum and forest products) main export products are more traditional.

Interestingly, three out of five in lowest performing countries show weakness in the severance pay (Croatia, Portugal, and Slovakia). Remaining two are weakest in population (Latvia and Slovenia). In strengths these countries seldom have common nominator – labour costs for LCI are strength of Latvia and Croatia, where Slovakia and Slovenia share high performance in foreign trade. As main export groups these countries have some similarities – automotive sector of Portugal, Slovakia and Slovenia, and petroleum products of Croatia and Portugal. Latvia is interesting mixture of higher technology (transmission apparatus) and wood. It could be said that these lower performing countries are

Table 2 Evaluation results of eleven evaluation factors for manufacturing unit location decision in EU-27 countries

	Foreign trade from GDP	GDP growth	GDP per capita	Labour force participation rate	Inflation	Population	Surface area	Labour cost for LCI	LPI index	Tax rate	Recent inflation	Labour productivity	Severance pay	Dependency ratio	Total
Ireland	0.061	0.096	0.075	0.037	0.048	0.011	0.017	0.019	0.037	0.056	0.046	0.271	0.004	0.047	0.826
France	0.017	0.025	0.044	0.037	0.040	0.152	0.133	0.017	0.040	0.026	0.064	0.037	0.008	0.034	0.674
Germany	0.024	0.026	0.050	0.034	0.033	0.186	0.086	0.017	0.044	0.023	0.044	0.040	0.004	0.033	0.646
Sweden	0.024	0.035	0.060	0.033	0.040	0.023	0.109	0.017	0.043	0.034	0.031	0.047	0.096	0.043	0.633
Italy	0.016	0.014	0.036	0.042	0.041	0.132	0.073	0.021	0.039	0.029	0.033	0.031	0.096	0.028	0.630
Spain	0.018	0.021	0.031	0.036	0.039	0.106	0.122	0.027	0.040	0.028	0.066	0.029	0.003	0.032	0.599
Luxembourg	0.098	0.040	0.126	0.038	0.033	0.001	0.001	0.015	0.038	0.028	0.070	0.043	0.010	0.046	0.587
Finland	0.021	0.023	0.052	0.035	0.038	0.012	0.082	0.018	0.042	0.035	0.042	0.044	0.096	0.036	0.575
Romania	0.023	0.048	0.012	0.040	0.024	0.043	0.058	0.076	0.033	0.043	0.023	0.010	0.096	0.036	0.564
Malta	0.082	0.072	0.030	0.039	0.036	0.001	0.000	0.037	0.030	0.020	0.052	0.022	0.096	0.047	0.563
Denmark	0.029	0.034	0.065	0.034	0.040	0.013	0.010	0.014	0.042	0.032	0.044	0.065	0.096	0.037	0.555
Cyprus	0.040	0.033	0.030	0.037	0.067	0.002	0.002	0.036	0.033	0.056	0.048	0.019	0.096	0.049	0.547
Belgium	0.045	0.029	0.050	0.039	0.031	0.026	0.007	0.015	0.043	0.028	0.037	0.062	0.096	0.038	0.544
Poland	0.027	0.050	0.016	0.039	0.029	0.084	0.075	0.057	0.037	0.037	0.023	0.016	0.007	0.039	0.536
Austria	0.029	0.025	0.053	0.035	0.029	0.020	0.020	0.017	0.042	0.028	0.037	0.044	0.096	0.036	0.512
Bulgaria	0.035	0.037	0.009	0.038	0.032	0.015	0.027	0.096	0.032	0.070	0.022	0.008	0.018	0.035	0.475
Hungary	0.046	0.044	0.016	0.040	0.023	0.022	0.022	0.063	0.036	0.077	0.015	0.017	0.007	0.040	0.469
Netherlands	0.043	0.029	0.055	0.034	0.030	0.039	0.010	0.017	0.042	0.027	0.039	0.054	0.006	0.035	0.461
Greece	0.020	0.010	0.021	0.039	0.083	0.024	0.032	0.037	0.034	0.032	0.053	0.015	0.004	0.030	0.433
Czechia	0.041	0.035	0.023	0.036	0.026	0.024	0.019	0.044	0.039	0.037	0.024	0.017	0.007	0.039	0.410
Lithuania	0.041	0.051	0.019	0.036	0.028	0.006	0.016	0.062	0.032	0.046	0.017	0.014	0.010	0.031	0.408
Estonia	0.043	0.050	0.023	0.035	0.027	0.003	0.011	0.046	0.035	0.035	0.022	0.017	0.018	0.038	0.400
Slovakia	0.051	0.037	0.020	0.038	0.031	0.012	0.012	0.046	0.032	0.033	0.025	0.014	0.007	0.041	0.398
Slovenia	0.042	0.037	0.026	0.037	0.038	0.005	0.005	0.031	0.035	0.037	0.037	0.022	0.008	0.034	0.394
Portugal	0.023	0.022	0.024	0.036	0.044	0.023	0.022	0.040	0.038	0.033	0.039	0.015	0.005	0.029	0.394
Croatia	0.026	0.033	0.015	0.041	0.039	0.009	0.014	0.058	0.033	0.039	0.029	0.013	0.006	0.034	0.387
Latvia	0.035	0.045	0.018	0.035	0.032	0.004	0.016	0.059	0.030	0.035	0.018	0.013	0.007	0.033	0.378

missing in general pharmaceutical and medical industry exports (apart of Slovenia) as compared to high performing ones. However, developing pharmaceutical and medical industries takes long period of time as knowledge, information management and different export markets create their challenges in global supply chains (e.g., markets of Middle-East see Zighan et al. 2024).

It is also beneficial to further analyze analysis results of Table 2. It is illustrated in Fig. 2 the performance of Ireland, Sweden, and Latvia among different manufacturing location variables (arranged in descending order of performance of Ireland). It could be said that Irish strength is somehow associated in economy and trade, and favourable tax rate. Country is also having best dependency ratio of these countries. Interestingly, inflation in long-term and currently is having favourable performance. In situation of Sweden, strengths are clearly in the surface area size, non-existing severance pay and high GDP per capita. Lowest performing Latvia is having strengths within low labour cost and high GDP growth. Largest catch-up of Latvia to Ireland and Sweden is in labour productivity, which should receive attention. Productivity is multi-faced issue, and it does not only concern "efficiency", but also it is an issue of which industries are the most important, what kind of international companies are in the country and how inflation is currently managed.

Table 3 Detected main strength and weakness of each EU-27 country with information of main export product groups. Source (data of exports): World Bank (2023b)

Country	Strength	Weakness	Main Export Product Groups (2020)	
Austria	Severance pay	Labour cost for LCI	Human and animal blood; microbial cultures; tox	Other medicaments of mixed or unmixed products
Belgium	Severance pay	Surface area	Other medicaments of mixed or unmixed products	Vaccines for human medicine
Bulgaria	Labour cost for LCI	Labour productivity	Copper cathodes and sections of cathodes unwrou	Petroleum oils, etc, (excl. crude); preparation
Croatia	Labour cost for LCI	Severance pay	Petroleum oils and oils obtained from bituminou	Petroleum oils, etc, (excl. crude); preparation
Cyprus	Severance pay	Population	Cargo vessels nes and other vessels for the tra	Petroleum oils, etc, (excl. crude); preparation
Czechia	Labour cost for LCI	Severance pay	Transmission apparatus, for radioteleph incorpo	Automobiles with diesel engine displacing more
Denmark	Severance pay	Surface area	Other medicaments of mixed or unmixed products,	Electric generating sets, nes
Estonia	GDP growth	Population	Transmission apparatus, for radioteleph incorpo	Petroleum oils, etc, (excl. crude); preparation
Finland	Severance pay	Population	Petroleum oils, etc, (excl. crude); preparation	Multi-ply paper and paperboard, coated
France	Population	Severance pay	Aircraft nes of an unladen weight exceeding 15,	Other medicaments of mixed or unmixed products,
Germany	Population	Severance pay	Other medicaments of mixed or unmixed products,	Automobiles with reciprocating piston engine di
Greece	Inflation	Severance pay	Petroleum oils, etc, (excl. crude); preparation	Other medicaments of mixed or unmixed products,
Hungary	Tax rate	Severance pay	Automobiles with reciprocating piston engine di	Television receivers including video monitors a
Ireland	Labour productivity	Severance pay	Human and animal blood; microbial cultures; tox	Other medicaments of mixed or unmixed products,
Italy	Population	GDP growth	Other medicaments of mixed or unmixed products,	Petroleum oils, etc, (excl. crude); preparation
Latvia	Labour cost for LCI	Population	Transmission apparatus, for radioteleph incorpo	Coniferous wood sawn or chipped lengthwise, sli
Lithuania	Labour cost for LCI	Population	Petroleum oils, etc, (excl. crude); preparation	Composite diagnostic or laboratory reagents, ne
Luxembourg	GDP per capita	Surface area	Sections,H,i/nas,nfw than hot rlld,drawn/extrud	Sheet piling,i or s whether or not drilled/punc
Malta	Severance pay	Surface area	Monolithic integrated circuits, digital	Other medicaments of mixed or unmixed products,
Netherlands	GDP per capita	Severance pay	Petroleum oils, etc, (excl. crude); preparation	Transmission apparatus, for radioteleph incorpo
Poland	Population	Severance pay	Television receivers including video monitors a	Electric accumulators, nes
Portugal	Inflation	Severance pay	Petroleum oils, etc, (excl. crude); preparation	Automobiles with reciprocating piston engine di
Romania	Severance pay	Labour productivity	Boards, panels, including numerical control pan	Ignition wiring sets&oth wiring sets of a kind
Slovakia	Foreign trade from GDP	Severance pay	Automobiles with reciprocating piston engine di	Automobiles nes including gas turbine powered
Slovenia	Foreign trade from GDP	Population	Other medicaments of mixed or unmixed products,	Automobiles with reciprocating piston engine di
Spain	Surface area	Severance pay	Automobiles with reciprocating piston engine di	Petroleum oils, etc, (excl. crude); preparation
Sweden	Surface area	Labour cost for LCI	Other medicaments of mixed or unmixed products	Petroleum oils, etc, (excl. crude); preparation

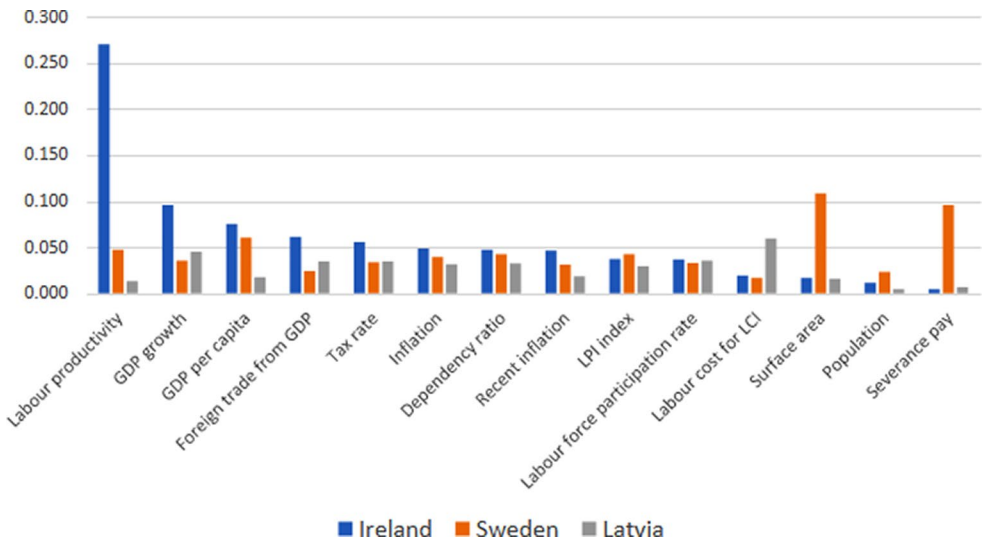


Fig. 2 Further performance analysis of Ireland, Sweden and Latvia in manufacturing location variables (in descending performance order of Ireland)

Strength of Ireland is apparent from comparison to two other high performing, but also to large European countries, France and Germany (Fig. 3). Ireland outperforms these two countries clearly in labour productivity, GDP growth, foreign trade from GDP and tax rate. However, as already noted earlier, larger European countries do hold their advantage. As Fig. 3 illustrates this, it is in geographical size and size of population.

As could be assumed, there is a tradeoff existing in between of labour productivity (score of evaluation, where higher is the better) and labour cost (score, where lower is better). Figure 4 illustrates that this scattergram could provide possibility to form clusters or groups from European countries, and they would be mostly based on east-west divide. However, what is striking, is the performance of Ireland, which is more than four times that of Denmark and Belgium in labour productivity (which are in the second and third place in productivity score). Irish labour cost is something between Finland and Italy, but it is clearly setting own standard in labour productivity. Based on Fig. 4, labour cost could increase rather significantly in Ireland to fit the tradeoff curve.

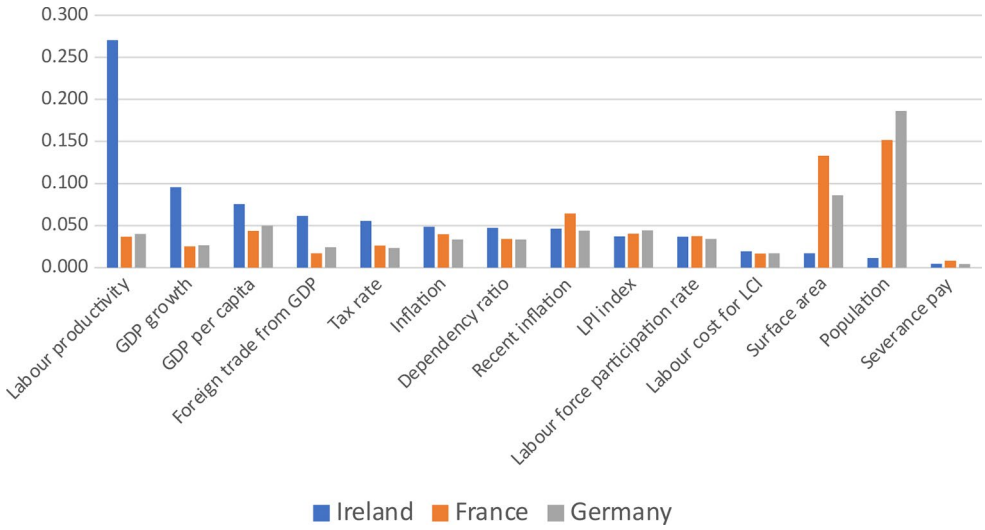


Fig. 3 Further performance analysis of Ireland, France and Germany in manufacturing location variables (in descending performance order of Ireland)

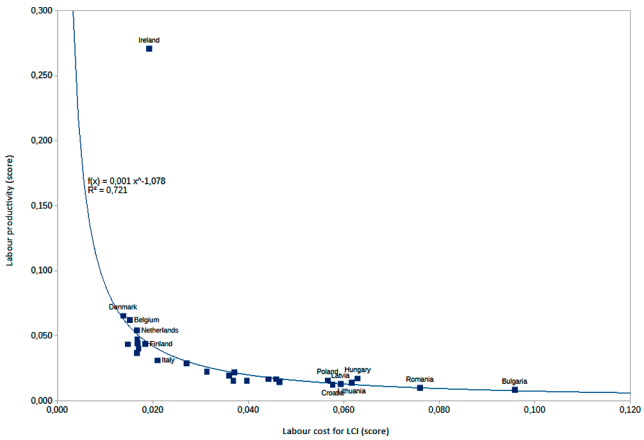


Fig. 4 Tradeoff in labour productivity and labour cost performance of evaluation

5 Rank-based evaluation

Earlier pairwise comparison method is in the following compared to classical rank-based evaluation. In here, all fourteen criteria are used to evaluate EU-27 countries, and highest performer gets rank of first, and next one second, and lowest performer will get 27th. As in some measures performance is in absolute terms the same between EU-27 countries, then same performance results in same rank (and lowest performer might have rank of 26, where e.g., two countries show lowest performance). In the following we have evaluated variables similarly as earlier, where in some preferred (and highest ranked) performance is highest performance (like in foreign trade from GDP, Population, Surface area etc.), but as earlier, it also could be lowest possible performance, which is desired most (like in Inflation, Severance pay, labour force participation rate etc.).

What is interesting in rank-based evaluation, is the finding that best performance could be detected from the same country as in pairwise evaluation, that of Ireland (Table 4). Country is also clearly leading EU-27 group of countries as it was in earlier evaluation too. However, after this we do not have two largest EU countries, but instead, Sweden. These two clearly high performing countries are followed by group of countries, which are having average rank of 12–13 (starting from Belgium in third position and ending to Romania, which is 13th). It should be noted that high performing large countries of pairwise comparison have considerably lost their positions as most preferred location. France is 11th, Italy is 12th, Germany is 17th and Spain is 18th. This could be explained with the fact that these countries are in some measures (e.g., in the size related measures) so much better performing that they gain excessive surplus as compared to others. In rank-based evaluation this does not exist as country only gets rank, and not measure how much better country is as compared to others.

As examining the lowest performing countries in rank-based list of Table 4, it could be noted that they are mostly the same smaller EU countries as they were in pairwise evaluation too. Lowest performer is Latvia, which was in the same position within pairwise comparison too.

Best performance (rank of 1) and lowest performance (rank of 26 or 27) was added to Table 4 to highlight performance countries in both lowest and highest performance positions. Ireland is having highest ranks in two dimensions: GDP growth and labour productivity. In turn, Sweden is having highest rank only in severance pay (sharing position with eight other countries as no severance pay is needed to be paid). Sweden is in turn lowest performing in one regard, which is labour force participation rate (it is in other words highest among EU-27 countries as lower participation rate was better). Rather interestingly, Luxemburg is in three measures as highest ranked in evaluation, but still only makes sixth in the evaluation of Table 4. This is

Table 4 Evaluation of EU-27 countries with manufacturing location variables and using the average rank of overall criteria

Rank	Country	Rank (aver.)	Rank 1 (highest)	Rank 26 or 27 (lowest)
1	Ireland	8.93	GDP growth (annual), Labour productivity	
2	Sweden	10.79	Severance pay	Labour force participation rate
3	Belgium	11.79	Severance pay	
4	Cyprus	12.00	Severance pay, Dependency ratio	
5	Denmark	12.14	Severance pay	Labour cost for LCI
6	Luxembourg	12.14	Foreign trade from GDP, GDP per capita, Recent inflation	
7	Finland	12.43	Severance pay	
8	Hungary	12.57	Tax rate	Recent inflation
9	Malta	12.57	Severance pay	LPI index, Population, Surface area, Tax rate
10	Poland	12.71		
11	France	12.93	Surface area	
12	Italy	12.93	Labour force participation rate, Severance pay	Dependency ratio, Foreign trade from GDP
13	Romania	13.36	Severance pay	Inflation
14	Austria	13.57	Severance pay	
15	Czechia	13.93		
16	Bulgaria	14.36	Labour cost for LCI	GDP per capita, Labour productivity
17	Germany	14.36	Population, LPI index	
18	Spain	14.43		Severance pay
19	Slovenia	14.57		
20	Netherlands	14.86		
21	Slovakia	14.93		
22	Estonia	15.79		
23	Greece	15.79	Inflation	GDP growth (annual)
24	Croatia	15.93		
25	Portugal	16.07		
26	Lithuania	16.43		
27	Latvia	17.79		

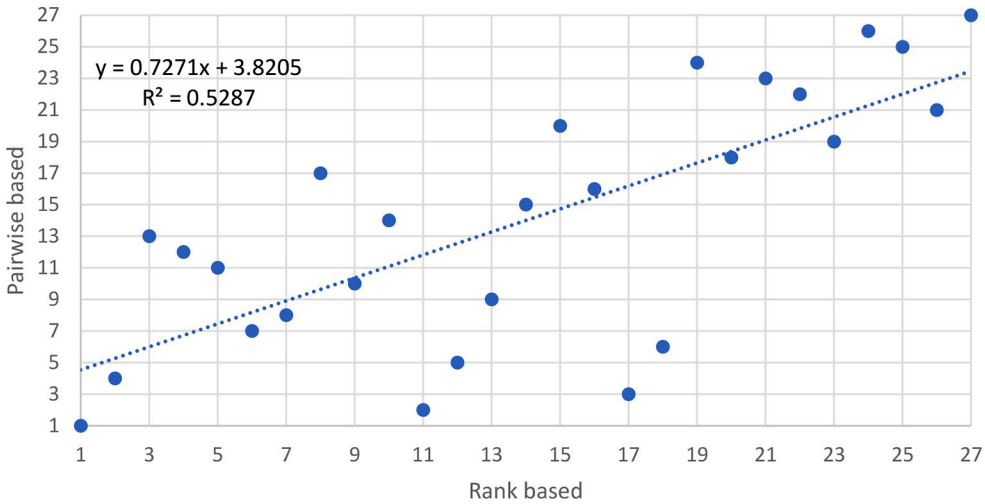


Fig. 5 Scattergram from pairwise and rank based evaluations of EU-27 countries

explained with performance of other variables, where country is having very low performance (not lowest) in population, surface area, labour costs and tax rate. Italy could also be considered as like Luxemburg as it is having highest ranks in two performance measures and lowest in two as well. Among lowest performing countries, we could say that they do not show highest or lowest ranks in any evaluation variable. Typically, they are below average or very low in their performance in all considered variables. However, all countries do show better performance in some measures, like Latvia being the fifth lowest labour cost country, and Lithuania being third highest in GDP growth.

In general, pairwise and rank based evaluations of EU-27 countries do give similar outcomes as Fig. 5 shows (rank-based performance is able to explain 52.87% from pairwise ranks – regression line is also having statistically significant linear coefficient). The difference between these two methods is in the treatment of largest countries as well as some small and mid-sized countries. In pairwise comparison method largest countries performed rather high as in rank-based Belgium, Cyprus and Denmark did get quite high ranks. In addition, Malta is having rather high overall performance in rank-based method (ninth), but is in pairwise method lower performing (17th).

6 Concluding discussion

In this study we have highlighted that member countries of the European Union exhibit notable differences that should be considered in manufacturing location decisions. The countries differ in several ways. Firstly, a pairwise general comparison model was applied and used second-hand data. It was not any industry specific model, where e.g., electricity and energy prices would have been one decision variable. Results were rather surprising as old West European countries were typically within top positions of the evaluation. It could be said that typically smaller and East European countries were lower preference locations. These could benchmark decisions and branches of e.g., Ireland to develop attractiveness of manufacturing industry further. Irish performance development in its manufacturing sector during the previous decade is of course learning point for all European countries – based on Eurostat (2023) its value added change was significantly better than any other European country in 2010–2020 (relative share), and it was in relative basis able to increase manufacturing employment most. This has been achieved in very demanding macroeconomic environment. In particular, labour productivity of Ireland is something, which ought also to gain attention among country and company level decision makers. However, even Ireland has its weaknesses. For example, severance pay is rather long, and in completed further analysis it was found that Sweden could act as learning point in this regard. Severance pay shortness was considered as an advantage of Denmark, Sweden and Finland among industrial participants of this study (taking part of joint-seminars).

Together with the performance variable benchmarking itself, it could be so that industrial structure is driving the performance. In this study it was found that pharmaceutical and medical industry plays often important role in higher performing countries. Secondly, the pairwise model results were compared to classical rank-based evaluation, where it could be concluded that dominance of Ireland and Sweden continued among top performing EU-27 countries (latter was also high performing in pairwise evaluation model). However, the largest EU-27 countries did not any longer make to highest performance levels. Overall, this study highlights the importance of considering the unique economic structures of EU member countries in manufacturing location decisions, as well as the crucial role played by certain industries in driving economic success.

In earlier research, manufacturing location evaluations have not provided that dramatic results as new challenging regions typically do not pose advantages of traditional locations (which used to be popular). In a previous study, Chou and Yu (2013) found international distribution center for case study company, and one location was clearly better than two other alternatives. Our conclusion is supported by Panova and Hilmola (2018), where inside of country one traditional and earlier very popular location was eventually challenged only by one new region. Findings of e-commerce location were rather same – in China traditional locations were competitive, and only one newcomer was challenging them (Panova et al. 2019). To our knowledge, no previous study has developed or conducted a similar type of analysis from European countries, yet the contextual and macro level factors are very relevant and of great importance regarding manufacturing relocation. In one way our findings are less dramatic as they emphasize the importance of larger EU countries as manufacturing facility location. However, some smaller countries do also make the list, and in many ways, Ireland could be considered as showing surprisingly high performance as it is located rather faraway from Central Europe and population as well as size of this country is small.

The findings of this study are of interest to managers as well as policy makers, in an environment called "win-lose" globalization by Mariotti (2024). The model provides in-depth analysis on aggregate macro level for each EU member country. This is of interest for practitioners, managers and firms who consider relocating their manufacturing and operations to Europe or want to develop an in-depth understanding of how the European markets differ. For policy makers, the study offers a novel comparison between member states to highlight the country differences and what policies can be implemented to offer more appealing location for foreign direct investments and manufacturing relocation. This macro level analysis is also an important tool not only for relocating manufacturing to Europe from other continents, but also to encourage manufacturing and firms to stay in the European Union. In order for firms to keep manufacturing in Europe, the macro level factors and conditions must be competitive over time.

With this study, we lay the foundation for future research. We would suggest building manufacturing location models for different industries. These could be used in the evaluation of key industries location decisions and provide improvement areas for countries lacking these industries. Case studies, surveys and/or focus groups discussions in the area are also sought after and would increase the understanding and criteria of location decision. These could reshape and adjust current evaluation model. In addition, researchers can investigate the manufacturing location decisions in other area, for example comparing manufacturing location decisions between EU countries and non-EU countries to identify differences and similarities.

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Declarations

Conflict of interest The authors declare no conflict of interest.

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